Borax is normally formulated as $Na_2B_4O_7.10H_2O$. But it contains, tetranuclear units $[B_4O_5.(OH)_4]^{2-}$. This form is known as prismatic form. Borax also exists two other forms namely, jeweller or octahderal borax ($Na_2B_4O_7.5H_2O$) and borax glass ($Na_2B_4O_7$).

Properties

Borax is basic in nature and its solution in hot-water is alkaline as it dissociates into boric acid and sodium hydroxide.

$$Na_2B_4O_7 + 7H_2O \longrightarrow 4H_3BO_3 + 2NaOH$$

On heating it forms a transparent borax beads.

$$Na_2B_4O_7.10H_2O \xrightarrow[-10H_2O]{\Delta} Na_2B_4O_7 \longrightarrow 2NaBO_2 + B_2O_3$$

Borax reacts with acids to form sparingly soluble boric acid.

$$Na_2B_4O_7 + 2HCl + 5H_2O \longrightarrow 4H_3BO_3 + 2NaCl$$

$$Na_2B_4O_7 + H_2SO_4 + 5H_2O \longrightarrow 4H_3BO_3 + Na_2SO_4$$

When treated with ammonium chloride it forms boron nitride.

$$Na_2B_4O_7 + 2NH_4Cl \longrightarrow 2NaCl + 2BN + B_2O_3 + 4H_2O$$

Uses of Borax:

1. Borax is used for the identification of coloured metal ions
2. In the manufacture optical and borosilicate glass, enamels and glazes for pottery
3. It is also used as a flux in metallurgy and also acts as a preservative

2.2.5. Boric acid [H_3BO_3 or $B(OH)_3$]:

Preparation:

Boric acid can be extracted from borax and colemanite.

$$Na_2B_4O_7 + H_2SO_4 + 5H_2O \longrightarrow 4H_3BO_3 + Na_2SO_4$$

$$Ca_2B_6O_{11} + 11H_2O + 4SO_2 \longrightarrow 2Ca(HSO_3)_2 + 6H_3BO_3$$

Properties:

Boric acid is a colourless transparent crystal. It is a very weak monobasic acid and, it accepts hydroxyl ion rather than donating proton.

$$B(OH)_3 + 2H_2O \rightleftharpoons H_3O^+ + [B(OH)_4]^-$$

It reacts with sodium hydroxide to form sodium metaborate and sodium tetraborate.

$$H_3BO_3 + NaOH \longrightarrow NaBO_2 + 2H_2O$$

$$4H_3BO_3 + 2NaOH \longrightarrow Na_2B_4O_7 + 7H_2O$$

Action of Heat:

Boric acid when heated at 373 K gives metaboric acid and at 413 K, it gives tetraboric acid. When heated at red hot, it gives boric anhydride which is a glassy mass.

XII_U2-P-Block.indd 34 22-12-2022 20:05:46